

Solutions to weekly problems and exercises: Week 7a

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Week 7a

1.2 Percentages and Ratio

Problem

Transform 3.6 into a percentage.

$$\frac{36}{10} = \frac{360}{100} = 360\%$$

Exercises

1. Transform 2.5 into a percentage.

$$\frac{25}{10} = \frac{250}{100} = 250\%$$

2. Calculate 12% of 60. 7.2
3. Calculate 11.4% of 100. 11.4

1.3 Calculating Percentages Changes

Problems

- Last month, 250 consumers bought a new smartphone. This month, the number is 420. What is the percentage increase in sale?

$$\frac{420-250}{250} \times 100\% = \frac{170}{250} \times 100\% = 68\%$$

- This year, your monthly rent is £750. Next year, the monthly rent is reduced by 5%. Calculate the new rent.

The new rent is $(100\% - 5\%) \times £750 = 95\% \times £750 = £712.5$

Exercises

1. Last month, 250 consumers bought a new smartphone. This month, the number is down to 100. What is the percentage decrease in sale?

$\frac{100-250}{250} \times 100\% = \frac{-150}{250} \times 100\% = -0.6 \times 100\% = -60\%$. In an absolute term, the percentage reduction is 60%.

2. Yesterday there were 250 unique visitors clicking a website. Today, you found that there is a 10% increase in visit. How many clicks are there today?

$$110\% \times 250 = 275$$